MOCKUP

User Stories:

1) As a Habitat Operator, I want to see the latest value of each sensor, so that I can monitor the current conditions inside the habitat.

2) As a Habitat Operator, I want sensor values to update automatically in real time, so that I can continuously monitor the habitat without refreshing the page.

3) As a Habitat Operator, I want to see the measurement unit of each sensor value, so that I can correctly interpret the data.

4) As a Habitat Operator, I want to see the name of each sensor, so that I can easily identify the device producing the data.

5) As a Habitat Operator, I want to see the timestamp of the latest sensor update, so that I can know how recent the displayed data is.

6) As a Habitat Operator, I want to monitor telemetry streams from habitat subsystems, so that I can observe continuously updated system metrics.

7) As a Habitat Operator, I want the dashboard to automatically receive telemetry data from streaming topics, so that telemetry metrics are updated in real time.

8) As a Habitat Operator, I want to visualize telemetry data such as power systems, radiation levels, and life support metrics, so that I can monitor critical habitat subsystems.

9) As a Habitat Operator, I want to visualize sensor and telemetry data in charts, so that I can observe trends while monitoring the system.

10) As a Habitat Operator, I want to see environmental indicators such as temperature, humidity, pressure, and air quality, so that I can ensure the habitat remains safe.

11) As a Habitat Operator, I want to create automation rules based on sensor values, so that the system can automatically react to dangerous conditions.

12) As a Habitat Operator, I want to define threshold conditions for sensors, so that actions are triggered when specific limits are exceeded.

13) As a Habitat Operator, I want to select which actuator should respond to a rule, so that the correct device is activated.

14) As a Habitat Operator, I want to enable or disable automation rules, so that I can control when automation is active.

15) As a Habitat Operator, I want to delete automation rules, so that I can remove rules that are no longer needed.

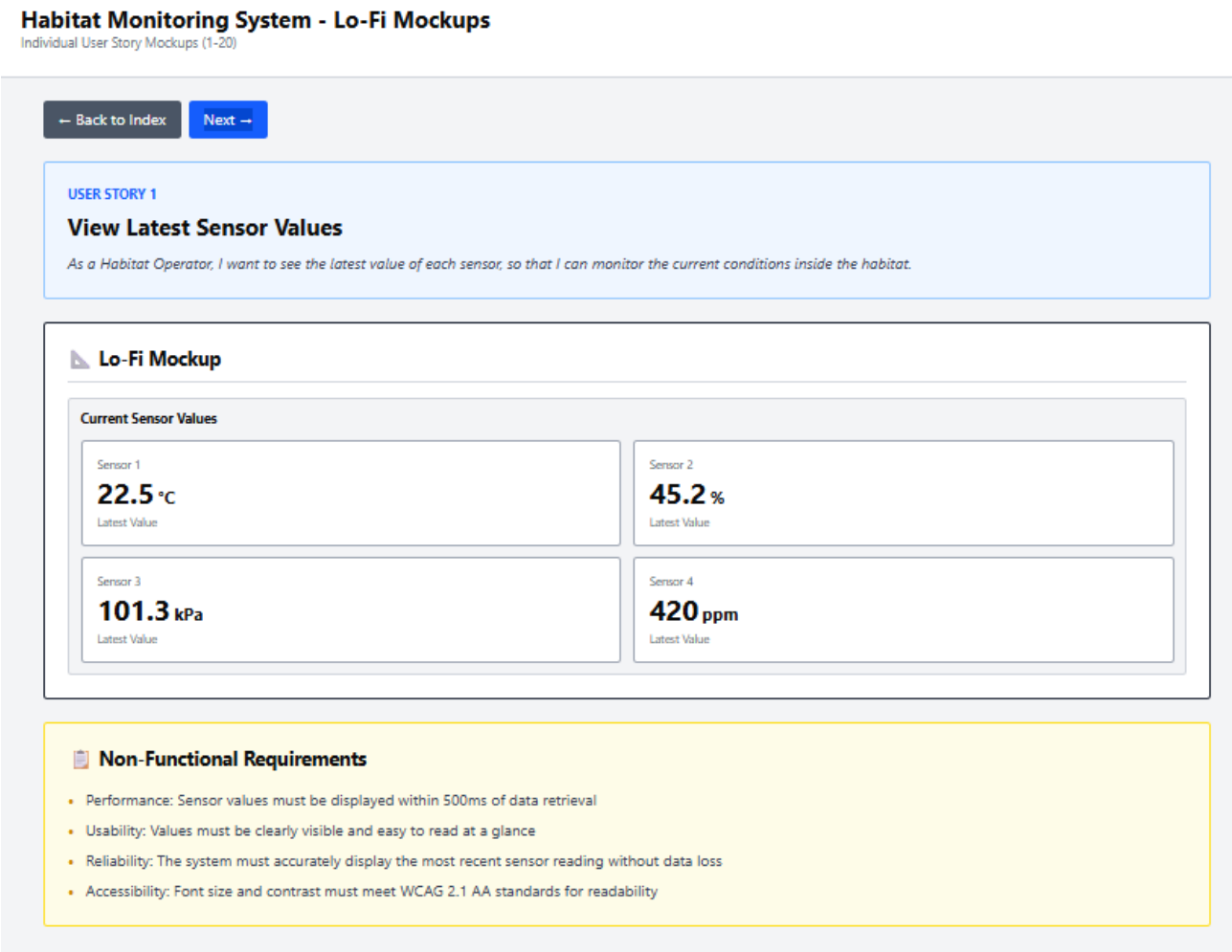
16) As a Habitat Operator, I want to see the list of all automation rules, so that I can understand the current automation logic of the system.

17) As a Habitat Operator, I want to see the condition and the action defined in each rule, so that I know how the system will react to sensor events.

18) As a Habitat operator I want to disable the rules of a certain actuator when it's manually turned on or off (the actuator).

19) As a Habitat Operator, I want to see the current state of each actuator, so that I know which devices are currently active in the habitat.

20) As a Habitat Operator, I want to receive alerts when sensor values exceed safe thresholds, so that I can intervene quickly if necessary.



Habitat Monitoring System - Lo-Fi Mockups

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USER STORY 2

Real-Time Auto-Updates

As a Habitat Operator, I want sensor values to update automatically in real time, so that I can continuously monitor the habitat without refreshing the page.

📐 Lo-Fi Mockup

✓ **Auto-Update Active (Updates every 2s)**

Update count: 1

Sensor 1

22.8 °C

Updating automatically...

Note: Values update automatically without page refresh. Watch the value change every 2 seconds.

📋 Non-Functional Requirements

- Performance: Updates must occur within 3 seconds of new sensor data availability
- Reliability: Auto-update mechanism must maintain 99.9% uptime
- Scalability: System must handle updates from up to 100 sensors simultaneously
- Efficiency: Updates must use minimal bandwidth (push/websocket preferred over polling)

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USER STORY 3

Display Measurement Units

As a Habitat Operator, I want to see the measurement unit of each sensor value, so that I can correctly interpret the data.

Lo-Fi Mockup

Sensor Values with Units

Sensor 1
22.5 °C
Unit: Degrees Celsius

Sensor 2
45.2 %
Unit: Percentage

Sensor 3
101.3 kPa
Unit: Kilopascals

Sensor 4
420 ppm
Unit: Parts per million

Note: Each measurement unit is clearly displayed next to its value for proper data interpretation.

Non-Functional Requirements

- Usability: Units must be displayed immediately adjacent to values to prevent misinterpretation
- Consistency: All units must follow international standard abbreviations (SI units preferred)
- Clarity: Units must be clearly distinguishable from numeric values through visual hierarchy
- Accuracy: Correct units must be displayed for each sensor type without error

Habitat Monitoring System - Lo-Fi Mockups

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USER STORY 4

Display Sensor Names

As a Habitat Operator, I want to see the name of each sensor, so that I can easily identify the device producing the data.

📐 Lo-Fi Mockup

Sensor Identification

📍 Sensor 1 **22.5** °C ID: s1	📍 Sensor 2 **45.2** % ID: s2
📍 Sensor 3 **101.3** kPa ID: s3	📍 Sensor 4 **420** ppm ID: s4

Note: Each sensor has a clear, identifiable name displayed prominently.

📋 Non-Functional Requirements

- Usability: Sensor names must be prominently displayed and easily readable
- Consistency: Naming convention must be standardized across all sensors
- Clarity: Names must uniquely identify each sensor to prevent confusion
- Persistence: Sensor names must remain consistent across sessions and updates

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USER STORY 5

Show Update Timestamp

As a Habitat Operator, I want to see the timestamp of the latest sensor update, so that I can know how recent the displayed data is.

Lo-Fi Mockup

Last Update:

17:40:21

10/03/2026

Sensor 1

22.5 °C

Updated: 10/03/2026, 17:40:21

Note: The timestamp updates every 5 seconds to show data freshness.

Non-Functional Requirements

- Accuracy: Timestamps must be accurate to within 1 second of actual update time
- Format: Timestamps must use a clear, human-readable format (ISO 8601 or locale-specific)
- Visibility: Timestamp must be prominently displayed and easy to locate
- Time Zone: System must clearly indicate the time zone or use UTC consistently

Habitat Monitoring System - Lo-Fi Mockups

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USER STORY 6

Monitor Telemetry Streams

As a Habitat Operator, I want to monitor telemetry streams from habitat subsystems, so that I can observe continuously updated system metrics.

Lo-Fi Mockup

Active Telemetry Streams

Telemetry 1	Telemetry 2	Telemetry 3
OK	OK	OK
126.94 kW	**4.75 cGy/h**	**20.92 %**
Streaming...	Streaming...	Streaming...

Note: Telemetry values stream continuously from subsystems.

Non-Functional Requirements

- Performance: Telemetry streams must update with latency under 2 seconds
- Reliability: Stream connections must auto-reconnect on failure within 5 seconds
- Scalability: System must support at least 50 concurrent telemetry streams
- Data Integrity: Streamed data must maintain accuracy with zero packet loss tolerance

Habitat Monitoring System - Lo-Fi Mockups

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USER STORY 7

Auto-Receive Streaming Data

As a Habitat Operator, I want the dashboard to automatically receive telemetry data from streaming topics, so that telemetry metrics are updated in real time.

📐 Lo-Fi Mockup

Stream Connection Status • Connected

Messages received: **1**

Subscribed Topics

- `habitat/telemetry/power` • Active
- `habitat/telemetry/radiation` • Active
- `habitat/telemetry/life-support` • Active

Note: Dashboard automatically subscribes to streaming topics on load.

📋 Non-Functional Requirements

- Architecture: Must use message broker (e.g., MQTT, Kafka) for streaming topics
- Performance: Message delivery latency must be under 500ms end-to-end
- Reliability: Must guarantee at-least-once message delivery
- Throughput: System must handle minimum 1000 messages/second

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USER STORY 8

Visualize Critical Subsystems

As a Habitat Operator, I want to visualize telemetry data such as power systems, radiation levels, and life support metrics, so that I can monitor critical habitat subsystems.

📐 Lo-Fi Mockup

Critical Habitat Subsystems

Power System — Nominal

126.94 kW

Radiation Monitor — Safe

4.75 cGy/h

Life Support — Operational

20.92 %

Note: Critical subsystems are monitored with status indicators.

📋 Non-Functional Requirements

- Usability: Critical subsystems must be clearly differentiated with color coding
- Safety: Anomalous values must trigger immediate visual indicators
- Clarity: Each subsystem must display current status, value, and operational state
- Priority: Critical systems must be visually prioritized over non-critical metrics

Habitat Monitoring System - Lo-Fi Mockups

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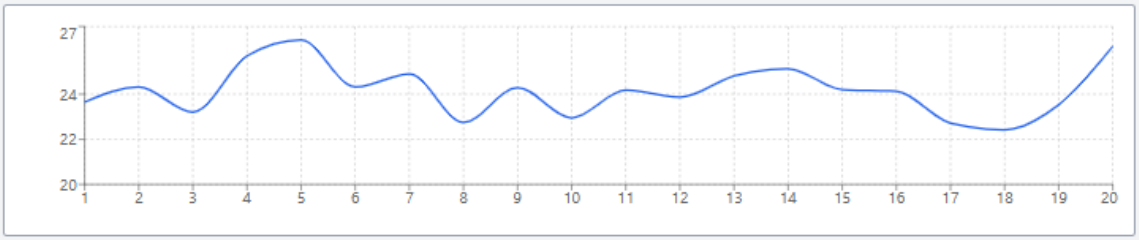
USER STORY 9

Visualize Trends in Charts

As a Habitat Operator, I want to visualize sensor and telemetry data in charts, so that I can observe trends while monitoring the system.

📐 Lo-Fi Mockup

Sensor 1 Trend (Last 20 readings)



Current Value	Trend
26.6 °C	↗ **Increasing**

Note: Charts update in real-time showing historical trends.

📋 Non-Functional Requirements

- Performance: Charts must render smoothly without lag (60 FPS minimum)
- Usability: Trend lines must be clearly distinguishable with appropriate colors
- Responsiveness: Charts must auto-scale based on screen size
- Data Density: Charts must display sufficient historical data (minimum 1 hour)

Habitat Monitoring System - Lo-Fi Mockups

Individual User Story Mockups (1-20)

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USER STORY 10

Environmental Indicators

As a Habitat Operator, I want to see environmental indicators such as temperature, humidity, pressure, and air quality, so that I can ensure the habitat remains safe.

Lo-Fi Mockup

Environmental Monitoring Dashboard

Temperature	Humidity	Pressure	Air Quality
Safe	Safe	Safe	Safe
22.5 °C	45.2 %	101.3 kPa	420 ppm
Min: 18 / Max: 26	Min: 30 / Max: 60	Min: 95 / Max: 105	Min: 300 / Max: 1000

✓ **Habitat Environment: SAFE**
All environmental parameters within safe ranges

Non-Functional Requirements

- Safety: Environmental values must be validated against safety thresholds
- Visibility: All critical environmental parameters must be visible on one screen
- Color Coding: Safe/Warning/Danger states must use universally recognized colors
- Response Time: Environmental changes must be reflected within 3 seconds

Habitat Monitoring System - Lo-Fi Mockups

Individual User Story Mockups (1-20)

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USER STORY 11

Create Automation Rules

As a Habitat Operator, I want to create automation rules based on sensor values, so that the system can automatically react to dangerous conditions.

📐 Lo-Fi Mockup

Automation Rules + Create New Rule

Note: Click the button to reveal the rule creation form.

📋 Non-Functional Requirements

- Usability: Rule creation interface must be intuitive with clear field labels
- Validation: All rule inputs must be validated before saving
- Performance: Rule creation must complete within 2 seconds
- Persistence: Created rules must be saved reliably to prevent data loss

Habitat Monitoring System - Lo-Fi Mockups

Individual User Story Mockups (1-20)

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USER STORY 12

Define Threshold Conditions

As a Habitat Operator, I want to define threshold conditions for sensors, so that actions are triggered when specific limits are exceeded.

Lo-Fi Mockup

Define Threshold Condition

Select Sensor

Sensor 1 (Temperature: 18-26 °C)

Condition

> (Greater than)

Threshold Value

26.0

Preview:

IF Sensor 1 > 26.0 °C

Non-Functional Requirements

- Precision: Threshold values must support decimal precision appropriate to sensor type
- Validation: Threshold values must be validated against sensor min/max ranges
- Flexibility: System must support multiple condition types (>, <, =, >=, <=)
- Clarity: Threshold configuration must clearly show units and acceptable ranges

Habitat Monitoring System - Lo-Fi Mockups

Individual User Story Mockups (1-20)

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USER STORY 13

Select Actuator Response

As a Habitat Operator, I want to select which actuator should respond to a rule, so that the correct device is activated.

Lo-Fi Mockup

Select Actuator & Action

Available Actuators

- **Actuator 1**
  Current State: OFF
- **Actuator 2**
  Current State: OFF
- **Actuator 3**
  Current State: OFF
- **Actuator 4**
  Current State: OFF

Action

Turn ON

Selected Action:
THEN **Turn ON** actuator **Actuator 1**

Non-Functional Requirements

- Usability: Available actuators must be clearly listed with descriptions
- Validation: Only compatible actuators should be selectable for given sensor types
- Safety: Critical actuators must require confirmation before selection
- Clarity: Each actuator must show its current state and capabilities

Habitat Monitoring System - Lo-Fi Mockups

Individual User Story Mockups (1-20)

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USER STORY 14

Enable/Disable Rules

As a Habitat Operator, I want to enable or disable automation rules, so that I can control when automation is active.

📐 Lo-Fi Mockup

Automation Rules

Rule 1
IF Sensor 1 > 26 THEN Turn ON Actuator 1
ENABLED | Disable

Rule 2
IF Sensor 1 > 26 THEN Turn ON Actuator 1
DISABLED | Enable

Rule 3
IF Sensor 1 > 26 THEN Turn ON Actuator 1
ENABLED | Disable

Note: Click buttons to toggle rules on/off.

📋 Non-Functional Requirements

- Responsiveness: Enable/disable action must take effect immediately (< 1 second)
- Visibility: Rule state (enabled/disabled) must be clearly indicated
- Safety: Disabling critical rules should show a warning message
- Audit: All enable/disable actions must be logged with timestamp and user

Habitat Monitoring System - Lo-Fi Mockups

Individual User Story Mockups (1-20)

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USER STORY 15

Delete Rules

As a Habitat Operator, I want to delete automation rules, so that I can remove rules that are no longer needed.

📐 Lo-Fi Mockup

Automation Rules

Rule 1
IF Sensor 1 > 26 THEN Turn ON Actuator 1
🗑 Delete

Rule 2
IF Sensor 1 > 26 THEN Turn ON Actuator 1
🗑 Delete

Rule 3
IF Sensor 1 > 26 THEN Turn ON Actuator 1
🗑 Delete

Note: Deletion requires confirmation to prevent accidental removal.

📋 Non-Functional Requirements

- Safety: Delete action must require explicit confirmation to prevent accidents
- Reversibility: Consider implementing soft delete with recovery option
- Audit: All deletions must be logged with timestamp and user information
- Performance: Delete operation must complete within 1 second

Habitat Monitoring System - Lo-Fi Mockups

Individual User Story Mockups (1-20)

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USER STORY 16

View All Rules

As a Habitat Operator, I want to see the list of all automation rules, so that I can understand the current automation logic of the system.

📐 Lo-Fi Mockup

All Automation Rules — Total: 3 | Active: 2

Rule 1 — ENABLED
Condition: Sensor 1 > 26
Action: Turn ON Actuator 1

Rule 2 — ENABLED
Condition: Sensor 2 < 30
Action: Turn ON Actuator 2

Rule 3 — DISABLED
Condition: Sensor 4 > 1000
Action: Turn ON Actuator 3

Note: Complete list of all configured automation rules.

📋 Non-Functional Requirements

- Completeness: All rules must be displayed with no pagination for critical systems
- Organization: Rules should be sortable by name, status, or creation date
- Clarity: Each rule must clearly show its condition, action, and current state
- Performance: Rule list must load within 2 seconds even with 100+ rules

Habitat Monitoring System - Lo-Fi Mockups

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USER STORY 17

See Condition and Action

As a Habitat Operator, I want to see the condition and the action defined in each rule, so that I know how the system will react to sensor events.

📐 Lo-Fi Mockup

Rule Details

Rule 1

CONDITION

IF **Sensor 1** > **26 °C**

ACTION

THEN **Turn ON** actuator **Actuator 1**

COMPLETE RULE

When **Sensor 1** reports a value **greater than 26 °C**, the system will automatically **turn ON Actuator 1**.

📋 Non-Functional Requirements

- Clarity: Condition and action must be displayed in plain, human-readable language
- Completeness: All relevant details (sensor, threshold, actuator) must be visible
- Format: Use consistent IF-THEN format for all rules
- Visibility: Condition and action should be visually distinct (e.g., color coding)

Habitat Monitoring System - Lo-Fi Mockups

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USER STORY 18

Disable Rules When Actuator Manually Controlled

As a Habitat operator I want to disable the rules of a certain actuator when it's manually turned on or off (the actuator)

📐 Lo-Fi Mockup

Manual Actuator Control

Actuator 1	Actuator 2	Actuator 3
ON	OFF	ON
Turn OFF / Turn ON	Turn OFF / Turn ON	Turn OFF / Turn ON

Automation Rules Status

Rule 1 (controls Actuator 1)	ENABLED
Rule 2 (controls Actuator 1)	ENABLED
Rule 3 (controls Actuator 2)	ENABLED
Rule 4 (controls Actuator 3)	ENABLED

💡 **How It Works**

1. When you manually click "Turn ON" or "Turn OFF" on an actuator
2. The system automatically disables ALL automation rules that control that actuator
3. This prevents automation from conflicting with your manual commands
4. You maintain full control until you re-enable the rules manually

Try it: Click any "Turn ON" or "Turn OFF" button above and watch the related rules automatically disable in the rules list below.

📋 Non-Functional Requirements

- Automation: Rules must be automatically disabled immediately upon manual actuator control
- Scope: Only rules associated with the manually controlled actuator should be disabled
- Notification: User must receive clear feedback about which rules were disabled
- Safety: Manual control must always override automation to ensure operator authority

Habitat Monitoring System - Lo-Fi Mockups

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USER STORY 19

View Actuator Current State

As a Habitat Operator, I want to see the current state of each actuator, so that I know which devices are currently active in the habitat.

📐 Lo-Fi Mockup

Actuator Status Dashboard

Actuator 1	Actuator 2
ON Currently Active	OFF Currently Inactive
Actuator 3	**Actuator 4**
OFF Currently Inactive	ON Currently Active

System Status: Active: 2 / Total: 4

📋 Non-Functional Requirements

- Real-time: Actuator state must update within 1 second of physical state change
- Clarity: State must be clearly indicated (ON/OFF) with visual distinction
- Reliability: State display must reflect actual device state with 99.9% accuracy
- Visibility: All actuators must be visible on one screen without scrolling

Habitat Monitoring System - Lo-Fi Mockups

Individual User Story Mockups (1-20)

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USER STORY 20

Receive Threshold Alerts

As a Habitat Operator, I want to receive alerts when sensor values exceed safe thresholds, so that I can intervene quickly if necessary.

Lo-Fi Mockup

CRITICAL ALERT
Sensor threshold violations detected

Active Alerts

Sensor 1 CRITICAL
Threshold Exceeded

Current Value:	Max Threshold:
28.5 °C	**26 °C**

Alert Time: 10/03/2026, 17:51:24

Acknowledge Alert

Note: Alerts appear immediately when thresholds are exceeded.

Non-Functional Requirements

- Timeliness: Alerts must be generated within 2 seconds of threshold violation
- Visibility: Alerts must be prominently displayed with high visual priority
- Severity: Different alert levels (warning/critical) must be clearly distinguished
- Persistence: Alerts must remain visible until explicitly acknowledged